MILITARY SPECIFICATION

INDICATOR, AIRSPEED, AND DIRECTION FIRE AND FLIGHT AIR DATA SUBSYSTEM, HELICOPTER ARMAMENT: M143

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains performance and environmental requirements, quality assurance provisions, and packaging for the Indicator, Airspeed and Direction, Fire and Flight Air Data Subsystem, Helicopter Armament: M143.

2. APPLICABLE DOCUMENTS.

2.1 Government documents. Unless otherwise specified the following Specifications and Standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	-	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	-	Electromagnetic Compatibility Requirements, Systems
MIL-I-8500	-	Interchangeability and Replacement of Component Parts for Aircraft and Missiles
MIL-F-13926	-	Fire Control Materials: General Specification Governing the Manufacture and Inspection of

FSC 1270

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-14806	-	Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	-	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-M-38510	-	Microcircuits, General Specification for
MIL-S-45743	-	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for
MIL-S-46844	-	Solder Bath, Soldering and Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

MILITARY

DOD-STD-100	-	Engineering Drawing Practices
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	-	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	-	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	-	Environmental Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, RESEARACH AND DEVELOPMENT CENTER (ARDC)

P11776300	-	Fire and Flight Air Data Subsystem M143

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11776301	-	Preservation and Packaging Data Sheet
11776301	-	Low Airspeed Indicator
11776302	-	Case Assembly
11776306	-	Chassis Assembly
11776309	-	Counter Balance Weight
11776322	-	Gear and Shaft Assembly
11776328	-	Cross Wire Assembly
11776337	-	Electronic Assembly
11776343	-	Lamp Assembly
11776549	-	Electronic Processor Unit
11776552	-	Processor, Signal Fire and Flight ADS M143
11776806	-	Detector, Wind Direction Speed and Temperature M143

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturer in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein:

American Society for Testing and Materials (ASTM).

ASTM-D3951	-	Standard Practice for Commercial Packaging.

2.1.3 Order of precedence. In the event of a conflict of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. All parts shall comply with the applicable drawings, specifications, and standards.

3.3 Air Data System Low Airspeed Indicator (LAI). The LAI shall comply with all requirements specified on Drawing (Dwg) 11776301, all associated drawings, and with all requirements specified in applicable specifications and standards. The LAI shall comprise a standard instrument case housing two servo mechanisms each driving a pointer across a circular dial.

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One pointer shall indicate airspeed from 50 knots forward to 50 knots aft and the other lateral airspeed from 50 knots left to 50 knots right. The LAI shall also display status GOOD for operation below 50 knots forward airspeed, status OVERRANGE for operation above 50 knots forward airspeed and status FAIL for an ADS or LAI failure condition or power interruption. The display surface shall be illuminated so as to be compatible with night vision goggles. The direction indicators shall be mounted on radius arms indirectly driven by torquer potentiometers. A printed circuit board shall contain the control electronics. The LAI shall be supplied with power, analog voltages and discrete voltage levels from the Electronic Processor Unit (EPU) drawing 11776549 in order that fore-aft and lateral airspeed in the range 0-50 knots may be displayed on the LAI dial. The discrete voltage levels are used to indicate ADS status and O/R (overrange) when the airspeed exceeds 50 knots.

3.3.1 Major component list.

11776302	-	Case Assembly
11776306	-	Chassis Assembly
11776328	-	Cross Wire Assembly
11776322	-	Gear and Shaft Assembly
11776337	-	Electronic Assembly
11776343	-	Lamp Assembly

3.4 Characteristics.

3.4.1 Electrical.

3.4.1.1 Signal interface. The LAI signal interface with the EPU shall be as defined in Appendix A to this specification and Figure 1.

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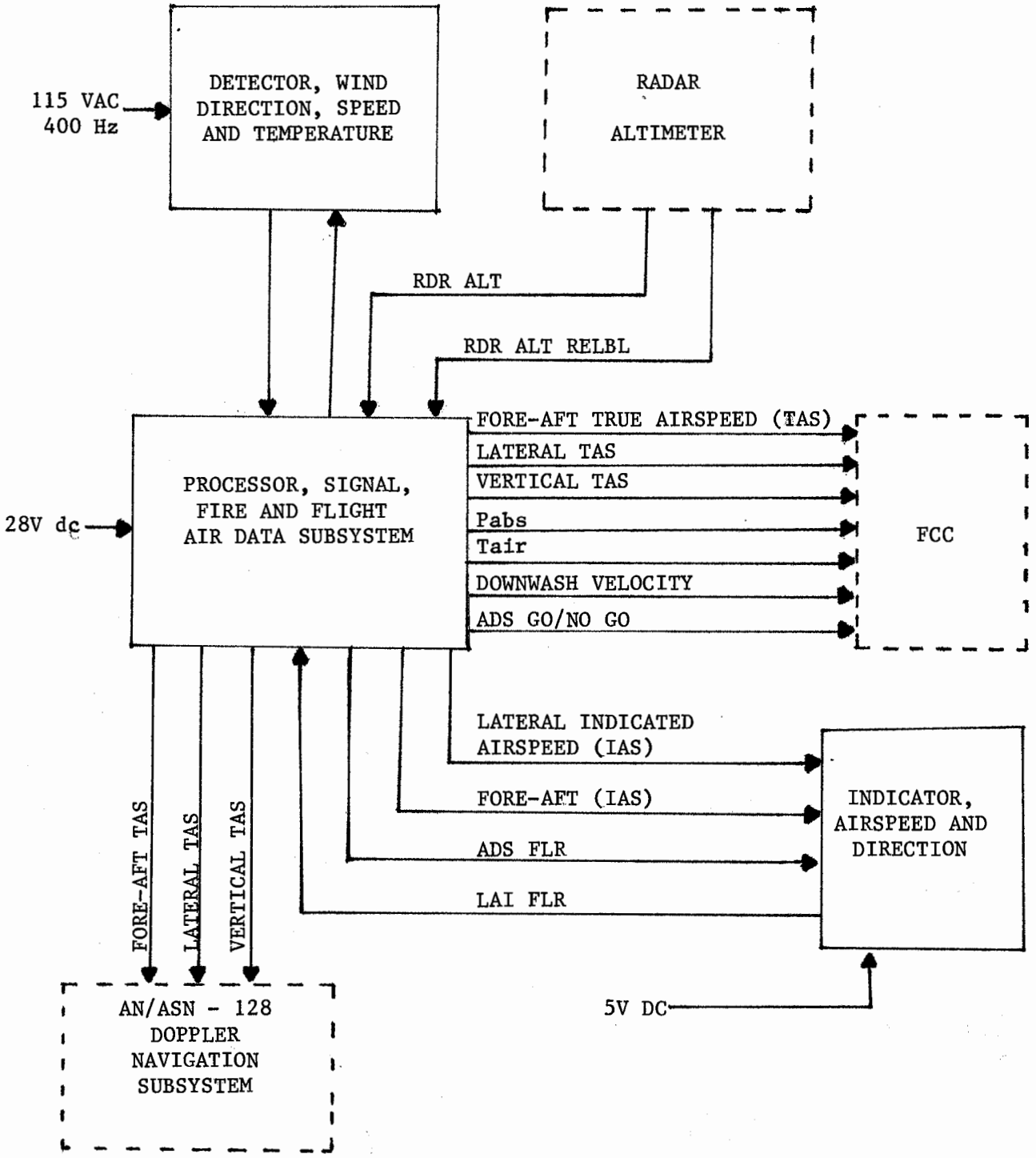


FIGURE 1. LAI interface block diagram

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3.4.1.2 Electrical power.

3.4.1.2.1 Steady state. The electrical service conditions will be in accordance with MIL-STD-704, Category "B" utilization equipment, except that the transient frequency limit of Figure 5, limit 4 in MIL-STD-704 shall be 240Hz. The equipment shall meet all specified requirements of MIL-STD-704 for the types of power specified herein, and shall give specified performance when supplied with electrical power having these characteristics and limits. The power required shall not exceed the amounts specified below:

a. 5V dc, 1A Panel Lighting Supply.

b. The LAI is supplied with ± 15 VDC power to a total of nine watts from the EPU.

3.4.1.2.2 Electrical power interruptions. Momentary power interruptions to the equipment will be experienced when external primary power is switched to helicopter power or when switching from one helicopter power bus to another. The LAI shall meet the performance requirements of this specification after such momentary power interruptions have been experienced. The LAI shall return automatically to specified performance within five seconds after any power interruptions.

3.4.1.3 Mechanical interface. The maximum envelope dimensions and mounting provisions of the LAI shall conform to Drawing 11776309.

3.4.1.4 Interfacing components. The LAI shall interface with the ADS EPU and the Instrument Lighting Control as per drawing 11776300.

3.4.2 Performance.

3.4.2.1 Operation. The LAI is used to display fore and aft and lateral airspeed and the status of the Air Data System (ADS) and operate in the range specified in Table I.

TABLE I. Indicated parameter range

Parameter	Units	Range	Tolerance
Lateral Indicated Airspeed	Knots	-50 to +50	±3 Kts (2σ)
Fore and Aft Indicated Airspeed	Knots	-50 to +50	±3 Kts (2σ)

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3.4.2.2 Input parameter. The LAI shall accept data in accordance with Table II.

TABLE II. Input parameter

Parameter	Type	Parameter Range	Voltage Range	Accuracy
Power	DC		± 15V	±1V
VIu	Analog	-50 to +50Kts	-5 to +5VDC	±3 Kts (2σ)
VIv	Analog	-50 to +50Kts	-5 to +5VDC	±3 Kts (2σ)
ADS Fail	Discrete	T or F	0 or +15VDC	

3.4.2.3 Output parameter. The LAI shall provide the following parameter to the EPU and display airspeed IAW Table III.

Parameter	Type	Range	Voltage
LAI Fail	Discrete	TorF	0 or 5 VDC

TABLE III. Output parameter

Parameter	Type	Parameter Range	Accuracy
VIu	Analog	-50 to +50 Kts	±3 Kts (2σ)
VIv	Analog	-50 to +50 Kts	±3 Kts (2σ)

3.4.2.4 Accuracy. The LAI accuracy values shown shall be defined as 95 per cent confidence values. LAI measured accuracy shall be within the specified accuracies in at least 19 out of each 20 separate measurements within the LAI performance envelope specified.

3.4.2.5 System turn-on time. The LAI response time from power-on actuation to LAI ready state shall not exceed five seconds.

3.4.2.6 Lighting. The LAI shall be provided with 5 volt red lighting conforming to MIL-L-25467.

3.4.3 Physical Characteristics.

3.4.3.1 Dimensions. Dimensional requirements applicable to envelope size, mounting and mating shall be in accordance with drawing 11776309.

3.4.3.2 Weight. The maximum weight of the LAI shall be 1 lb. 7 ozs.

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3.4.4 Preconditioning (burn-in). Each LAI procured under this specification shall be subjected to a burn-in test as defined in Figure 2.

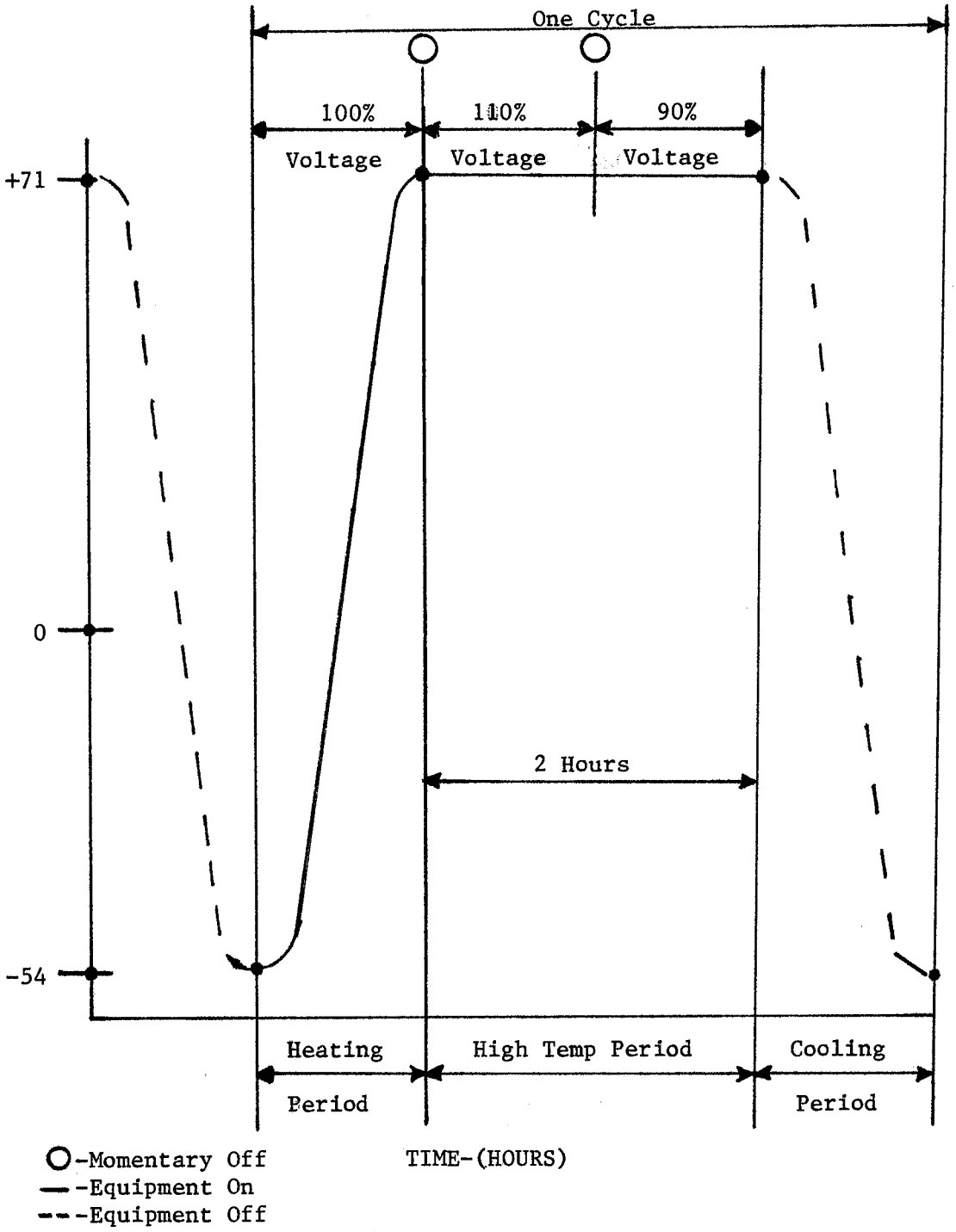


FIGURE 2. Typical temperature cycle - electronic and avionic equipment - continued

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3.4.5 Environmental conditions (first article testing only). The LAI shall operate and meet the performance requirements of 3.4.2 under any of the environmental conditions, or a reasonable combination of these conditions, as specified herein.

3.4.5.1 Natural environment. The LAI shall operate and withstand storage in the environments described in MIL-E-5400 except as specified herein.

3.4.5.1.1 Altitude. Operating, 0-20,000 feet; Nonoperating and storage 0-50,000 feet.

3.4.5.1.2 Temperature. Operating, nonoperating, and storage: -54° to $+71^{\circ}$C.

3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 509.1, Procedure II. In Step 6, measurements shall be taken during the last five hours of the last test cycle.

3.4.5.1.4 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the EPU shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

3.4.5.1.5 Fungus. No evidence of viable fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

3.4.5.1.6 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.1, Procedure I, except test to be limited to a maximum altitude of 20,000 feet.

3.4.5.1.7 Rain. Operation after being subjected to MIL-STD-810, Method 506.1, Procedure I. During each thirty minute rain period, the wind shall be maintained for fifteen minutes, commencing five minutes after the start of the period as specified in the procedures. The EPU shall be operated during the last ten minutes of the thirty minute period.

3.4.5.2 Service environment. The LAI shall withstand the service environment as specified in MIL-E-5400 for Class II equipment, except as noted in paragraphs 3.4.5.1.1 and 3.4.5.1.2.

3.4.5.2.1 Bench handling. The LAI shall withstand shock environments encountered in maintenance and servicing.

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3.4.5.3 Induced environment.

3.4.5.3.1 Vibration. The LAI shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Method 514.2, Procedure I, Part I, Table 514.2-III of MIL-STD-810; except that the frequency shall be limited to 500 Hz (reference Figure 3).

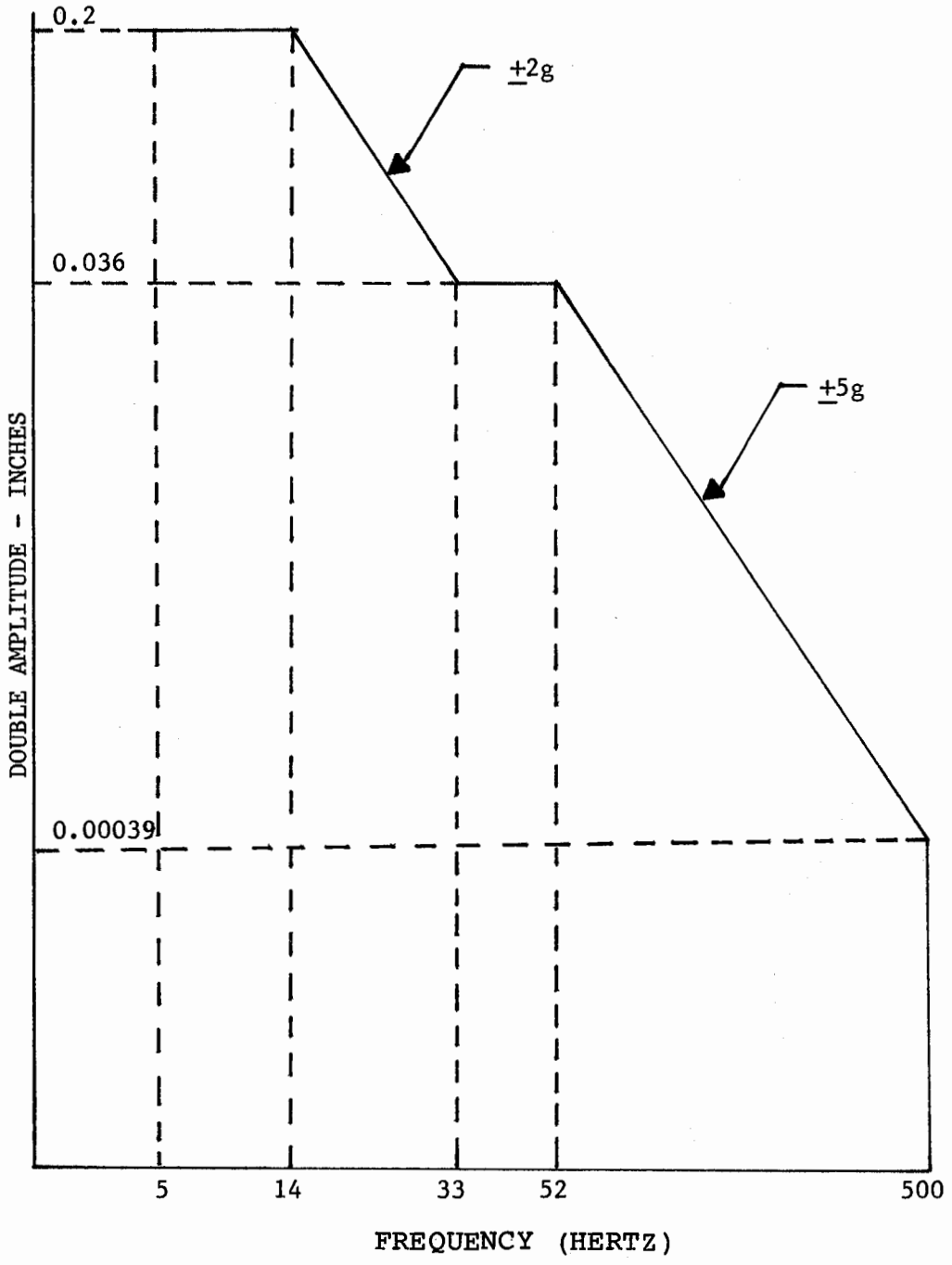


FIGURE 3. Vibration amplitude for qualification test

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3.4.5.3.2 Shock. The LAI shall meet the shock requirements of MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2 and Method 516.2, Procedure III.

3.4.6 Transportability. The LAI shall withstand vibration environments encountered during transportion as specified in MIL-STD-810, Method 514.2, Procedure X.

3.4.7 Processes.

3.4.7.1 Soldering. All soldering shall be performed in accordance with MIL-S-45743 or MIL-S-46844.

3.4.7.2 High efficiency anti-reflection (HEA) coating. All reflecting glass surfaces of the LAI shall be provided with a HEA coating conforming to MIL-C-14806.

3.4.7.3 Microelectronic devices. Microelectronic devices shall conform to MIL-M-38510.

3.4.8 Electromagnetic radiation.

3.4.8.1 Electromagnetic compatibility. Each LAI as installed in the helicopter as part of the Air Data Subsystem, shall not cause undesireable response, malfunction, or degraded performance to any other equipment/subsystem installed in or associated with the helicopter; nor shall any component malfunction, exhibit degraded performance, or shall undesireable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the helicopter are singularly or collectively operated and performing their intended functions.

3.4.8.2 Electromagnetic interference. The LAI when operated as part of the Air Data Subsystem shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE01, CE03, CE04, CE05, CS01, CS02, CS06, RE02, RS03, RE03.1, and RS02.

3.4.8.3 Circuit isolation. Signal and power potentials shall be electrically isolated from each other and the equipment chassis.

3.4.9 Interchangeability (First Article only). All components having the same manufacturer's part number shall be dimensionally and functionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part number and changes thereto. Interchangeability must be maintained within the detail parts to the level of replaceable items.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

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3.6 Workmanship. Workmanship shall be in accordance with MIL-F-13926 and electronic and mechanical weapon-build practices to ensure safety, proper and dependable operation, and service life. Particular attention shall be given to assure thoroughness of assembly, alignment of parts, electrical connections, cleanliness, tightness of screws and bolts, marking of parts, application of protective finish, and freedom from burrs and rough or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926.

4.4 First article The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three LAI's selected at random by the Government representative from the first fifteen production LAI's.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Tables IV and V.

4.4.3 Failure. Failure of any LAI to meet the requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any LAI to comply with any stated requirement.

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TABLE IV. First article tests

ITEM	CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
1.	Natural environment	3.4.5.1	4.6.4.1
2.	Altitude	3.4.5.1.1	4.6.4.2
3.	Temperature	3.4.5.1.2	4.6.4.3
4.	Humidity	3.4.5.1.3	4.6.4.4
5.	Salt atmosphere	3.4.5.1.4	4.6.4.5
6.	Fungus	3.4.5.1.5	4.6.4.6
7.	Explosive atmoshpere	3.4.5.1.6	4.6.4.7
8.	Rain	3.4.5.1.7	4.6.4.8
9.	Bench Handling	3.4.5.2.1	4.6.4.9.1
10.	Vibration	3.4.5.3.1	4.4.4.10.1
11.	Shock	3.4.5.3.2	4.6.4.10.2
12.	Transportability	3.4.6	4.6.6.10.3
13.	Electromagnetic radiation	3.4.8	4.6.4.11

TABLE V. Performance tests

ITEM	CHARACTERISTICS	REQUIREMENT	TEST PROCEDURES
101	VI_u Fore-Aft Pointer	3.4.2.3	4.6.2.3
102	VI_v Lateral Pointer	3.4.2.3	4.6.2.3

TABLE VI. General tests

ITEM	CHARACTERISTICS	REQUIREMENT	TEST PROCEDURES
301	Fabrication and General Specifications	3.1, 3.2, & 3.3	Applicable Drawings, Visual, and MIL-E-5400, Visual

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each LAI in every lot shall be subjected to the tests in Table V and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

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4.5.1.2 Acceptance. Where any LAI fails to meet any specified requirement, the defective LAI shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One LAI, as a control sample, shall be selected at random by the Government representative from each 100 LAI's produced and shall be subjected to the tests in Table VI.

4.5.2.2 Acceptance. Where any LAI fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in Table VII shall be used where specified to perform the required tests.

TABLE VII. Test equipment

Item/Nomenclature	Standard	Peculiar	Manufacturer & Model or Part No.
1. Test set ADS (LRU)		X	11779002
2. Digital Voltmeter (DVM)	X		Fluke 800 or equivalent
3. Ohmeter	X		Solartrom 7055 or 7075 or equivalent capable of measuring 0.1 ohms
4. Insulation Resistance Tester 100V	X		Hewlett Packard HP4329A or equivalent
5. LAI Lighting Comparison Standard		X	Photo Research Model UBD Spectra Spotmeter or equiv.
6. Variable DC Power Supply 0-10 Vdc, 0.5 Amps	X		Optional
7. Stopwatch accurate to 0.1 sec	X		Optional
8. Weighing machine capable of weighing 1.5 lbs.	X		Optional

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4.6.2 Performance tests. The LAI should be connected to the Test Set, ADS (LRU) by the appropriate cable harness as shown in Figure 4.

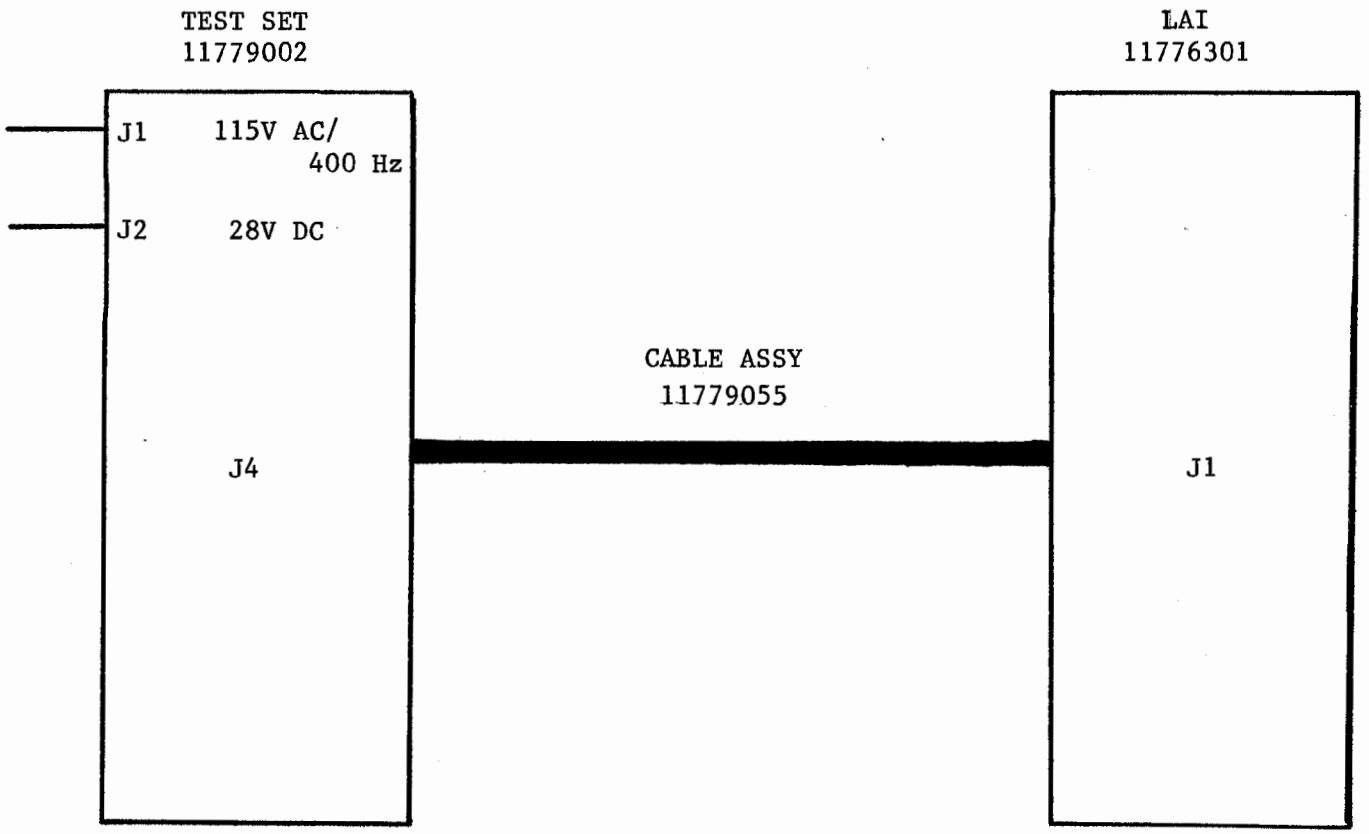


FIGURE 4. Test set ADS (LRU) - LAI test connection

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4.6.2.1 Turn-on time. Set S19 and S20 on the Test Set ADS (LRU) to 0 KTS. Set S21 on the Test Set ADS (LRU) to ON and observe the stopwatch and the LAI status flag; the flag shall show a black face within five seconds of operating S21.

4.6.2.2 Current test. Set S19 and S20 on the Test Set ADS (LRU) to 0 KTS. A multimeter, set to read dc current, shall be connected between the terminals marked AMPS (Red +ve) on the Test Set ADS (LRU). Set S13 on the Test Set ADS (LRU), to LAI +15V and depress S15. With the pointers stationary, the multimeter shall indicate a current of less than 300mA. Set S13 on the Test Set ADS (LRU) to LAI -15V and reverse the multimeter connections to the Test Set, then depress S16. With the pointers stationary, the multimeter shall indicate a current of less than 65mA.

4.6.2.3 Calibration. All dials on the LAI TEST panel on the Test Set ADS (LRU) shall be set to zero. Both pointers shall read zero ±2 knots when viewed head on at a distance of 18 inches from the instrument face. The tests which follows shall be viewed in a similar manner.

NOTE: Zero ±2 knots shall be checked by reference to Figure 5.

Rotate switch S20V_{iv}KTS to the +20 position. The lateral pointer shall display a reading of 20 ±2 knots, to the right.
Rotate switch S20V_{iv}KTS to the +50 position. The lateral pointer shall display a reading of 50 ±3 knots, to the right.
Rotate switch S20 V_{iv} KTS to the -20 position. The lateral pointer shall display a reading of -20 ±2 knots, to the left.
Rotate switch S20 V_{iv} KTS to the -50 position. The lateral pointer shall display a reading of -50 ±3 knots, to the left.
Set switch S20 V_{iv} KTS to zero.
Rotate switch S19 V_{iu} KTS to the -20 position. The Fore-Aft pointer shall display a reading of -20 ±2 knots, down.
Rotate switch S19 V_{iu} KTS to the -50 position. The Fore-Aft pointer shall display a reading of -50 ±3 knots, down.
Rotate switch S19 V_{iu} KTS to the +20 position. The Fore-Aft pointer shall display a reading of +20 ±2 knots, up.
Rotate switch S19V_{iu} KTS to the +45 position. The Fore-Aft pointer shall display a reading of +45 ±3 knots, up.
Reset switch S19V_{iu} KTS to zero.
Both pointers shall read zero knots ±2 knots.

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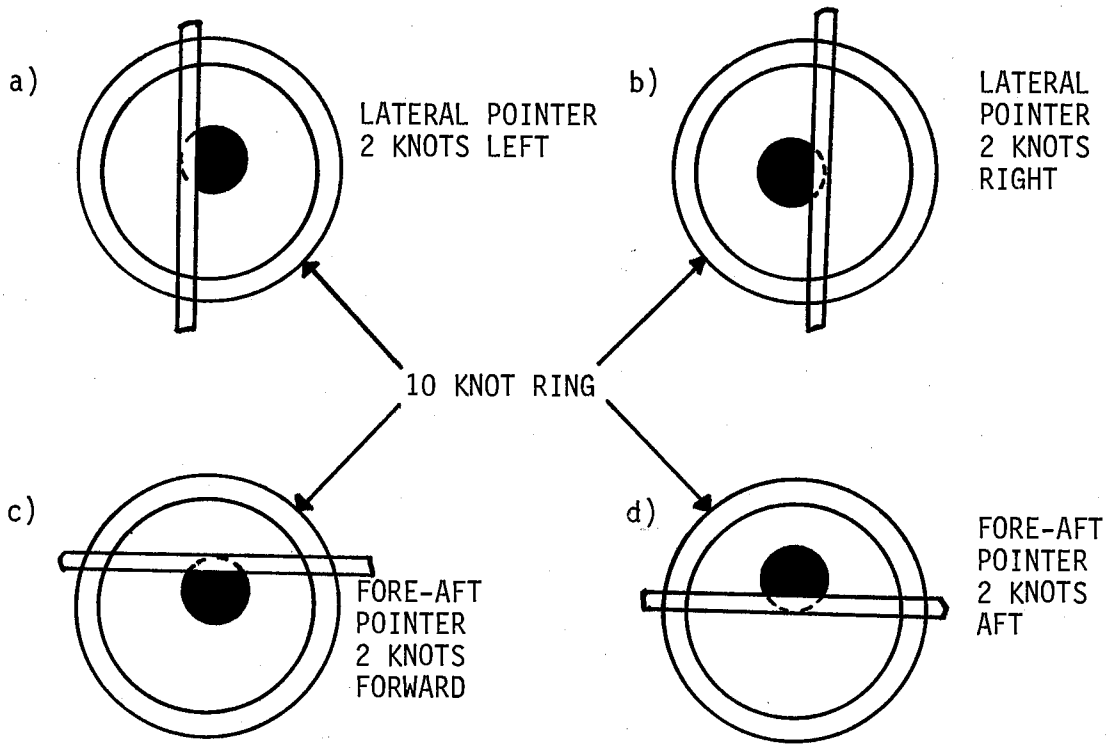


FIGURE 5. LAI pointers at 2 knots

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4.6.2.4 Failure mode. The LAI shall be connected for test as shown in Figure 3. Set switch S21 to OFF (DOWN), the status flag shall show a failure condition indicated by red and black diagonal stripes. Set switches S21 and S22 to ON (UP). The LAI status flag and the LAI "FAIL" indicator on the Test Set ADS (LRU) shall indicate a failure condition when each of the following operations are carried out:

Set switch S20 V_{iv} KTS to -50 position, then rotate to the position marked OUT LEFT. Reset to zero when the operation is complete, i.e. after the LAI "FAIL" indicator on the test set has illuminated.

Set switch S20 V_{iv} KTS to the +50 position, then rotate to the position marked OUT RIGHT. Reset to zero when the operation is complete.

Set switch S29 V_{iu} KTS to the -50 position. Then rotate to the position marked OUT AFT. Reset to zero when the operation is complete.

Set the switches S19 V_{iu} KTS and S20 V_{iv} KTS to zero and depress the ADS "FAIL" switch. The status flag on the LAI shall indicate a fail condition. Release the ADS "FAIL" switch on the Test Set.

4.6.2.5 Over-range mode. The status flag shall indicate O/R (over-range) when the following operation is carried out on the Test Set ADS (LRU).

Set switch S20 V_{iv} KTS to zero. Rotate switch S19 V_{iu} KTS to the +45 position. The status flag shall show O/R when switch S19 V_{iu} KTS is rotated to the +51 position.

Reset the switches to zero when the operation is completed.

4.6.2.6 Operational mode. When the LAI is connected for test as shown in Figure 4, and S19 and S20 are set to 0 KTS, the status flag shall show operational mode, indicated by a plain black display.

4.6.2.7 Panel lighting. The LAI and Lighting Comparison Standard shall be connected as shown in Figures 4 and 6.

Depress swith S23 (PNL Lights).

Set the stabilized supply to 4.10 $\pm$ .05V and compare the two instrument faces in dark conditions; no part of the face under test shall be less bright than the equivalent part of the standard face.

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Set the stabilized supply to 6.10 ± .05V and compare the two instrument faces-no part of the face under test shall be brighter than the equivalent part of the standard face.

Switch S21, S22, and S23 to their OFF positions. Disconnect the unit.

4.6.3 Physical characteristics.

4.6.3.1 Dimensions. Use standard practices to ensure the requirements of drawings are met.

4.6.3.2 Preconditioning (burn-in). Subject each LAI to the burn-in of Appendix B.

4.6.3.1.2 Weight. Use item 8 of Table VII to ensure that requirements of 3.4.3.2 are met.

4.6.4 Environmental tests.

4.6.4.1 Natural environment. Subject the EPU to the following environments.

4.6.4.2 Altitude. the LAI shall be subjected to the tests of Method 504.1 of MIL-STD-810 for category 4 equipment. The LAI nonoperating, shall be subjected to Method 500.1, Procedure I, with pressures simulating air transport at 50,000 feet altitude at a pressure of 2.85 inches of mercury (hg) for one hour. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.3 Temperature. The LAI shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810, except Step 4 temperature shall be 71°C. The system shall then be subjected to the test of Method 502.1, Procedure I, except maximum low temperature shall be -54°C. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.4 Humidity. The LAI shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810. In step 6, measurements shall be taken during the last five hours of the last test cycle. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.5 Salt atmosphere. The LAI shall be subjected to the test Method of 509.1 of MIL-STD-810. Operation of the LAI shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

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4.6.4 Fungus. The EPU shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces.

4.6.4.7 Explosive atmosphere. The LAI shall be subjected to test Method 511.1, Procedure I, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table V.

4.6.4.8 Rain. The LAI shall be subjected to test Method 506.1, Procedure I, of MIL-STD-810. During each thirty minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedures. The system shall be operated during the last 10 minutes of the thirty minute period. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.9 Service environment.

4.6.4.9.1 Bench handling. The system shall be subjected to Method 516.2, Procedure V, of MIL-STD-810. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.10 Induced environment.

4.6.4.10.1 Vibration. The LAI shall be subjected to test Method 514.2, Procedure I, Part I, Table 514.2-III, of MIL-STD-810, except that the frequency shall be limited to 500 cycles (reference Figure 3). At the conclusion of this test, subject the LAI to the performance test of Table V.

4.6.4.10.2 Shock. The LAI shall be subjected to:

a. Method 516.2 of MIL-STD-810, Procedure I, Figure 516.2-2 for flight vehicle equipment. The test "g" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter. The pulse test level and duration are defined in Paragraph 3.2.24.6.1 of MIL-E-5400.

b. Method 516.2, Procedure III for mounting base crash safety. The pulse test level and duration are defined in Paragraph 3.2.24.6.2 of MIL-E-5400. At the conclusion of this test, subject the LAI to the performance test of Table V.

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4.6.4.10.3 Transportability. The system shall be subject to the test of Method 514.2, Procedure X, of MIL-STD-810. At the conclusion of this test, subject the LAI to the performance tests of Table V.

4.6.4.11 Electromagnetic radiation. The LAI shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461 for the requirements of 3.4.8.

5. PACKAGING

5.1 Preservation, packing, packaging, and marking. The LAI shall be preserved, packaged, packed, and marked in accordance with the requirements of the preservation and packaging data sheet P11776301 at the level of protection specified in the contract (see 6.2b).

5.2 Commercial packaging shall be in accordance with ASTM-D3951.

6. NOTES

6.1 Intended use. This equipment is intended for use in the AH-1s (modernized) helicopters.

6.2 Ordering data.

6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number, and data of this specification.

b. Applicable packaging Data Sheet (see 5.1).

c. Selection for applicable levels of preservation, packaging and packing.

d. Applicable stock number.

e. First Article inspection requirements.

6.3 Previous revisions of MIL-I-70485 were for the US Army Armament Research and Development Center internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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Appendix A

Connector Identification and Interface

Connector Type: MS3122-E14-19P

Pin	Nomenclature	Cross Reference
A	Signal Ground (DV Connected to EPU)	EPU
B	LAI-15V Power	Ident 2
C	Panel Lighting Return	Ident 1
D	Panel Lighting +5V	Ident 1
E	Spare	
F	Fore-Aft Indicated Airspeed (VI_u)	Ident 3
G	Fore-Aft Indicated Airspeed RTN	Ident 3
H	Lateral Indicated Airspeed (VI_v)	Ident 4
J	Lateral Indicated Airspeed RTN	Ident 4
K	LAI Failure (LAI FLR/BAR)	Ident 5
L	ADS Failure (ADS FLR/BAR)	Ident 6
M	LAI +15V Power	Ident 2
N	Spare	
P	Spare	
R	Spare	
S	Spare	
T	Spare	
U	Spare	
V	Spare	

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Appendix A

Signal Interface Requirements Ident Number 1

Power Signal Interface Requirements

Nomenclature	LAI ±15V
Type	DC
Conductor Type and Guage	Twisted Shielded Pair 22 AWG
Origin	EPU
Destination	LAI
Destination Impedence	N/A
Source	EPU
Source Impedence	N/A
Power Required	N/A
Phase	N/A
Maximum Current (Total)	600 mA
Nominal Voltage	±15VDC
Maximum Voltage	±16VDC
Minimum Voltage	±14VDC
Power Factor	N/A
Regulation	N/A

Remarks LAI Power

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Appendix A

Signal Interface Requirements Ident Number 4

Discrete Signal Interface Requirements	
Nomenclature	LAI Failure (LAI FLR/BAR)
Type	Discrete
Conductor Type and Guage	22 AWG Single Unshielded
Origin	LAI
Destination	EPU
Destination Impedence	$\geq$1K ohm
Source	LAI
Source Impedence	True State $>$10K ohms
	False State $<$25 ohms
Maximum Current	5mA
Maximum Voltage	+5 VDC
True State Voltage	Open Circuit[1]
False State Voltage	DV $\pm$0.2V

Remarks

[1]Signal to be pulled up to +5V by EPU.

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Appendix A

Signal Interface Requirements Ident Number 5

Discrete Signal Interface Requirements	
Nomenclature	ADS Failure (ADS FLR/BAR)
Type	Discrete
Conductor Type and Guage	22 AWG Single Unshielded
Origin	EPU
Destination	LAI
Destination Impedence	>1K ohm
Source	EPU
Source Impedence	True State >10K ohms
	False State <25 ohms
Maximum Current	5mA
Maximum Voltage	+15 VDC
True State Voltage	Open Circuit[1]
False State Voltage	DV ±0.2V

Remarks

[1]Signal is pulled up to +15V by LAI.

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Appendix A

Signal Interface Requirements Ident Number 2

Nomenclature	Fore-Aft Indicated Airspeed (VI_u)
Type	DC Analog
Conductor Type and Guage	Twisted Shielded Pair 22 AWG
Origin	EPU
Destination	LAI
Destination Impedence	>10K ohms
Source	EPU
Source Impedence	<300 ohms
Voltage Range	-5.0 to +5.0 VDC
Scale Factor	100 Millivolts per knot
Parameter Range	-50 to +50 knots*
Parameter Accuracy	±4 knots (2σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	-5.0 VDC to +5.0 VDC

Remarks

*Positive voltage corresponds to Headwind/Forward flight.

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Appendix A

Signal Interface Requirements Ident Number 3

Analog Signal Interface Requirements	
Nomenclature	Lateral Indicated Airspeed (VI_V)
Type	DC Analog
Conductor Type and Guage	Twisted Shielded Pair 22 AWG
Origin	EPU
Destination	LAI
Destination Impedence	$\geqslant$10K ohms
Source	EPU
Source Impedence	$\leqslant$300 ohms
Voltage Range	-5.0 to +5.0 VDC
Scale Factor	100 Millivolts per knot
Parameter Range	-50 to +50 knots*
Parameter Accuracy	$\pm$4 knots (2σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	+5 VDC

Remarks

*Positive voltage corresponds to Rightward flight or sidewind from the right.

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Appendix A

Power Signal Interface Requirements	Ident Number 133
Nomenclature	LAI Lighting Power and Return
Type and Frequency	DC
Conductor Type and Guage	22 AWG
Origin	
Destination	LAI
Destination Impedence	N/A
Source	Pilot's Station Controls
Source Impedence	
Power Required	5 Watts
Phase	N/A
Maximum Current	1 AMP
Nominal Voltage	5 Volts, Dimmable
Maximum Voltage	5 VDC
Minimum Voltage	0 VDC
Power Factor	N/A
Regulation	N/A

Remarks

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Appendix B

BURN-IN REQUIREMENTS

Burn-in is to be applied to all deliverable systems or LRU's.

Test Environments

Environmental exposure for the LAI during burn-in is Level F of MIL-STD-781B. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5°C per minute. Figure 2 illustrates one cycle of the environmental profile described below.

Temperature	-54°C to +71°C (-65°F to 160°F).
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G $\pm$ 10% peak acceleration value at 30 $\pm$ 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling	A. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. B. Momentary off. C. 100% of nominal for next 1/3 of "ON" time. D. Momentary off. E. 90% of nominal for last 1/3 of "ON" time.

NOTE: Voltage Cycling will occur on the primary LAI power (28 VDC) only.

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Appendix B

Number of Cycles

The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

Duty Cycle Information

During equipment ON time, mode selection time shall be apportioned as follows:

Test	95% ± 5%

During operation the display and BIT Indicators shall be monitored.

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